Customer Segmentation & Churn Pattern Analytics in European Banking

Research Paper | Exploratory Data Analysis and Retention Recommendations

Prepared from the supplied European_Bank.csv dataset | October 2026

Abstract

This study examines customer churn in a supplied dataset of 10,000 banking customer records. It uses descriptive exploratory data analysis and segmentation by geography, age, gender, activity and product count. The observed overall churn rate is 20.37% (2,037 records). Germany, customers aged 46–60, inactive members and customers with three or four products show elevated observed churn in this sample. These patterns can help prioritize further investigation, but do not establish causes or represent all European banking customers.

1. Introduction

Customer churn can reduce customer lifetime value and increase the cost of acquiring replacement customers. Aggregate churn metrics alone may hide meaningful differences between customer groups. Segment-based analysis can help retention teams identify where to investigate service quality, engagement, product fit and customer needs. This project provides a reproducible, interactive descriptive analysis of the supplied dataset.

2. Research Questions

• What is the overall observed churn rate?

• How does churn vary across geography and demographic segments?

• How are engagement, tenure and number of products associated with churn?

• What is the observed churn profile of customers with higher account balances?

3. Dataset and Preparation

The supplied CSV contains 10,000 rows and 14 columns, including customer identifiers, credit score, geography, gender, age, tenure, balance, product count, credit-card ownership, activity status, estimated salary and the binary target `Exited`. No missing cells were detected in the initial file inspection. `Exited=1` is treated as churn and `Exited=0` as retained. Surname is not used as an analytical feature. Age groups are Under 30, 30–45, 46–60 and Over 60. Credit score bands are Low (≤579), Medium (580–669) and High (≥670). Tenure groups are 0–2, 3–6 and 7–10 years.

4. Methodology

The workflow loads the CSV, validates expected columns, checks missingness and duplicates, derives segmentation variables, calculates segment sizes and churn rates, and presents results using Streamlit and Plotly. Segment churn rate is calculated as the mean of the binary `Exited` field multiplied by 100. The high-balance explorer uses the 75th percentile of balance in the current filtered sample as a relative proxy. This definition is transparent but should not be confused with a bank-defined premium tier or customer profitability.

5. Results

Overall churn was 20.37% (2,037 churned; 7,963 retained). Geography-level rates were Germany 32.44% (814/2509), France 16.15% and Spain 16.67%. By age band, observed churn was Under 30 7.56%, 30–45 15.30%, 46–60 51.12% and Over 60 24.78%. Inactive members had 26.85% churn versus 14.27% among active members. Churn was 25.07% for female records and 16.46% for male records. Product-count churn rates were 1 product(s): 27.71% (n=5084); 2 product(s): 7.58% (n=4590); 3 product(s): 82.71% (n=266); 4 product(s): 100.00% (n=60). Among churned records, mean balance was 91,108.54 and mean estimated salary was 101,465.68.

6. Discussion

The largest geographic difference in this file is Germany's higher churn rate. The 46–60 age segment has the highest rate among the defined age groups. Inactivity is associated with a higher churn rate, making engagement a practical area for deeper investigation. The 3- and 4-product groups have high observed churn but contain only 266 and 60 records respectively; rates for small groups are less stable and require validation. Balance and salary averages alone do not establish financial value or revenue risk.

7. Recommendations

• Prioritize a focused diagnostic review of the Germany segment, including service journeys, product mix, customer feedback and potential data-quality differences.

• Design ethical engagement experiments for inactive customers and evaluate results using control groups where feasible.

• Review the needs of customers aged 46–60 through service and product-fit research; do not use age as the sole basis for adverse treatment.

• Investigate the small 3- and 4-product groups individually before drawing conclusions; validate whether these records reflect unusual product configurations or sample artifacts.

• Build a validated customer-value measure that accounts for margin, product profitability, cost-to-serve and lifetime value rather than balance alone.

• Monitor segment churn over time and assess whether interventions reduce churn without unfairly disadvantaging protected or vulnerable groups.

8. Limitations and Ethics

The data's provenance and sampling process were not independently verified. The dataset may be a benchmark or synthetic sample and should not be treated as representative of all European banks. This is descriptive analysis, not a causal study or a predictive model. Segment associations can be confounded by unobserved variables. Small subgroups can yield unstable estimates. Customer-level data should be protected, and decisions should be audited for fairness and compliance.

9. Conclusion

Segmentation reveals important differences hidden by the overall churn rate. In this dataset, churn is higher in Germany, among customers aged 46–60, among inactive members and in some small product-count groups. The dashboard enables users to explore these patterns interactively. The appropriate next step is targeted investigation and controlled retention experiments, not automatic customer decisions based on descriptive rates.

References

• Dataset: European_Bank.csv supplied with the project brief; source and provenance were not independently verified.

• Pandas documentation: https://pandas.pydata.org/docs/

• Streamlit documentation: https://docs.streamlit.io/

• Plotly Python documentation: https://plotly.com/python/